

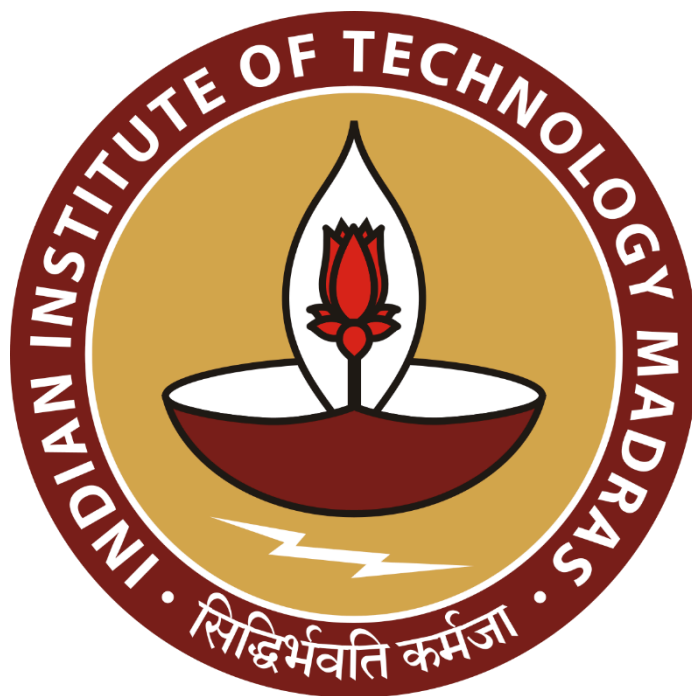
“Optimizing Inventory Management and Profitability for Sudha Milk Parlor”

A Final Report for the BDM Capstone Project

Submitted by

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1 Executive Summary

Sudha Milk Parlor is a state co-operative dairy shop located in Bihar, that has been selling dairy products for the past 19 years. Although the shop has managed to generate a decent level of profit during its period of operation, it has struggled to further enhance its profitability due to two major challenges: inefficient utilization of the inventory space and excessive wastage of products. The project was undertaken to address these two crucial issues in the business and ultimately improve the overall profitability of the shop.

In order to address these problems, thorough analyses were performed on the sales and purchase data of the shop. The main objectives of these analyses were to identify best-selling and slow-moving products, evaluate the sales conversion rates of individual products and calculate the profitability of each product. These analyses were essential to identify which products need to be prioritized over the other products.

Based on the results of these analyses, clear recommendations are proposed regarding the optimal purchase quantity of each product to better manage inventory levels, reduce investment in loss-incurring products and scale up the purchase of high-performing products. In addition to this proposal, a dynamic pricing strategy is also proposed as a contingency plan for products that are nearing expiration to minimize wastage.

By implementing these proposed data-driven recommendations, the shop can enhance its profitability by reallocating its resources towards fast-moving, profitable products, while minimizing losses from unsold or slow-moving products. In addition, the overall wastage of products can also be significantly reduced and potentially converted into revenue.

2 Detailed Explanation of Analysis Process/Method

This section provides a brief overview of the shop, outlines the different challenges associated with it, states the core objectives of the project, highlights a brief explanation of the analyses performed on the data to identify solutions for these challenges and the appropriate software tools that were used to carry out the analyses.

2.1 Organization Background

Sudha Milk Parlor is a dairy shop located in Hajipur, Vaishali, Bihar, and is owned by Mr. Vasant Kumar Singh. The shop follows a B2C (Business-to-Consumer) business model. For the past 19 years, it has been catering to the needs of the local community. The shop operates even during weekends and remains closed only on major festivals, thus being available to customers almost every day. The shop sells three different milk products to the customers, namely Sudha Gold Milk, Shakti Milk and Cow Milk. Each of these products is available in two different quantity variants: 500mL and 1L. These milk products differ in terms of the fat

content they possess, with Sudha Gold Milk having the highest fat percentage (6%), followed by Shakti Milk (4.5%) and Cow Milk having the least amount of fat content (3%).

Despite being a small-scale outlet, the shop has consistently attracted a good number of customers and generated a considerable amount of revenue over the years. However, its profitability has remained stagnant and has not shown significant growth in recent years. This has significantly hindered the shop's growth and profitability.

2.2 Challenges Faced by Sudha Milk Parlor

The following factors have significantly affected the shop's profitability and have contributed to considerable financial losses:

- **Inefficient Utilization of Inventory Space**

Being a small-scale shop, it is essential to maximise the use of limited inventory space to achieve higher profit margins. However, the shop has struggled in this regard. Overstocking of slow-moving products has led to inefficient inventory space usage and considerable financial losses.

- **Product Wastage**

Due to the short shelf-life nature of dairy products, it is crucial to ensure that the products are sold within a short period of time after procurement. However, non-optimal purchasing of slow-moving products has resulted in a significant number of products being unsold and thus, being discarded after the corresponding expiration date. These slow-moving products not only result in wastage, but also occupy a considerable part of the limited inventory space, which could have been filled with the other best-selling products.

- **Lack of Data-Driven Decision-Making**

This factor is the root cause of the above listed challenges faced by the shop. The shop lacks a structured, data-dependent strategy for purchasing products to tackle the challenges associated with inventory space utilisation and product wastage. The product-wise profitability is not analysed for individual products, and a fixed quantity of each product is typically purchased every time regardless of the sales performance.

These challenges highlight the need for a data-dependent approach to monitor the importance of individual products that are sold in the shop and accordingly modify the purchase decisions in the future. It is highly important to implement such an approach to not only monitor the sales performance of individual products, but also to improve the profit levels of the shop by making wise investments while procuring products.

2.3 Objectives of the Project

The ultimate goal of this project is to improve the profitability of Sudha Milk Parlor and reduce product wastage in the shop by using data-driven insights. In order to achieve this primary goal, the following sub-objectives were established:

- **Identification of best-selling and slow-moving products**

Categorizing products based on their sales performance can help distinguish those products which can contribute to the shop's revenue from those that hinder its overall profitability. This classification is important to understand which products should be procured in large quantities and which products are supposed to be procured in lesser numbers.

- **Recommendation of Inventory Optimization Strategies**

The optimal purchase quantities of each product can be analysed based on the findings from the above sub-objective, which can be implemented to avoid overstocking of slow-moving products. Such a strategy is important to ensure that a large part of the investment is directed towards the best-selling products while purchasing products.

- **Proposal of a Dynamic Pricing Model**

In order to tackle the challenge of expired or unsold products, a dynamic pricing strategy can be implemented as a contingency plan. This enables the sale of near-expiry products at substantially lower prices, thus minimizing product wastage and potentially increasing revenue.

NOTE: The impact of the implementation of a dynamic pricing model on the shop's profitability is not analyzed in the project, as throughout the course of the data collection period, only fixed pricing strategy was used in the shop. Hence, the implementation of a dynamic pricing strategy is only provided as one of the recommendations to the shop. By implementing such a strategy, it is suggested that possible product wastage can be avoided and on top of that, additional revenue can also be generated.

2.4 Methodology

The following subsections highlight the different analyses performed on the data to tackle the challenges presented in the problem statements associated with the shop and come up with feasible solutions for solving the same.

2.4.1 Data Collection and Preparation

Since the project emphasises the importance of data-driven decision-making, comprehensive and relevant data was collected from the shop owner to support the analysis. The data was collected in the form of [handwritten logs](#) on a daily basis. The collected data played a crucial role in evaluating the sales performance and profitability of individual products sold at Sudha Milk Parlor. The video interaction with the shop owner can be found [here](#).

In order to study the sales trends of individual products sold by the shop and identify best-selling versus slow-moving products, daily sales records of the shop were collected from the shop owner for a duration of three months - from December 2024 to February 2025.

These records provided the following key information:

- Quantities of each product sold per day
- Revenue generated from each product per day
- Overall revenue generated each day

In addition to the sales data, monthly purchase records of the shop were also collected from the shop owner, to better understand the inventory efficiency and sales conversion of each product. This supplementary dataset included:

- Quantity of each product purchased during the same three-month period.
- Cost incurred for procurement (i.e) total purchase price of each product

Once the above-mentioned data was collected, it was converted into digital format by storing it in Excel spreadsheets to perform the corresponding analysis. There were a few inconsistencies observed in the data, mainly in the product naming (different names used for the same product in some records; For example, in certain records, it was “SM1” and in certain records, it was “Shakti 1”) and numerical entries (unclear handwriting; For example, in certain records, it was difficult to differentiate the number “3” from “8”). Since these inconsistencies were results of human error, these were clarified through direct consultation with the shop owner to ensure accuracy of the overall dataset. The dataset comprising the sales data and purchase data of the shop can be accessed by clicking [here](#).

2.4.2 Analysis

To derive accurate insights from the dataset, several key calculations were performed during the analyses. These include revenue contribution, profit margin estimation, sales conversion rate, and optimal purchase quantity determination of individual products. The following is a detailed explanation of each of these metrics:

- **Product-wise Revenue Contribution**

The amount of revenue generated from the sale of each product was analyzed to identify which products are the major contributors to the shop's overall revenue and which products contribute the least to the profitability. For each milk product, the total revenue was computed by summing up the daily revenues across the 3-month period. This was then divided by the overall revenue to determine each product's contribution percentage:

$$\text{Revenue Contribution (\%)} = (\text{Total Revenue of Product} / \text{Total Revenue of All Products}) * 100$$

This analysis helped in prioritizing the high-performing products from the rest of the products.

- **Sales Conversion Rates**

Sales conversion rate was calculated by comparing the quantity of each product sold to the quantity purchased by the shop.

$$\text{Sales Conversion Rate (\%)} = (\text{Quantity Sold} / \text{Quantity Purchased}) * 100$$

This gave clear insights into which products are fast-selling and which products are slow-moving. Higher conversion rates indicated fast-moving products, while lower conversion rates pointed to inefficiencies. Identifying products with low conversion rates is crucial for recognizing products that occupy the limited inventory space for extended periods of time and are more likely to result in wastage.

- **Profit Margins**

Profit Margins for individual products were calculated by comparing their overall purchase cost from the total sales revenue generated during the period.

$$\text{Profit Margin (₹)} = \text{Total Revenue} - (\text{Quantity Purchased} * \text{Purchase Price per Unit})$$

This analysis helped in identifying which products contributed the most to the shop's overall profitability and which products resulted in minimal (or negative) returns effectively.

- **Optimal Purchase Quantity**

The optimal purchase quantity for each product was calculated based on two key parameters:

- Average daily quantity sold of each product.
- Sales conversion rate of each product

These parameters are used to estimate how much of each product should ideally be restocked every 5 days (which is the restocking frequency followed by the shop). Additionally, a small safety margin of 5% (represented as “Buffer” in the below formulation) was included during the calculation process to buffer against any minor demand fluctuations.

The mathematical formulation used to calculate the individual product’s optimal purchase quantity is:

$$\text{Optimal Quantity} = \text{Average Daily Sales} * \text{Restocking Period} * \text{Conversion Rate} * \text{Buffer}$$

The basic idea behind this estimation was to ensure that the fast-selling products (with high conversion rates and consistent demand) are purchased in sufficient quantities, while slow-moving products are stocked more conservatively. This prevents overstocking of less sold products and helps utilize the limited inventory space more effectively.

2.4.3 Software Tools Used

To carry out the appropriate analyses and visualizations on the dataset, the following software tools and programming environments were utilized:

- **Microsoft Excel**

Microsoft Excel was used as the first step in the data pipeline to digitize the handwritten sales and purchase records provided by the shop owner. These records were entered into structured tabular formats with appropriate column names, ensuring data integrity and uniformity across the dataset. Excel was also used to perform basic calculations such as daily revenue, product totals, and descriptive statistics. Additionally, initial insights like formatting inconsistencies and resolving product name ambiguities were handled using Excel’s data validation and filtering features.

- **Python Programming Language**

Python served as the primary tool for conducting in-depth analysis and data visualization on the dataset. The following libraries were employed:

- **Pandas:** It was used extensively for data manipulation, aggregation, and transformation. Pandas enabled smooth handling of the data, grouped summaries by product, and derivation of custom columns such as profit, conversion rates, and daily averages.

- **Matplotlib and Seaborn:** These libraries were used to generate visualizations like line plots, pie charts, bar graphs, and grouped comparisons. These made interpretation easier through well-labeled plots and color-coded distinctions between different categories.

This integration of Excel for structuring and Python for deep analysis allowed for a robust workflow that ensured both accuracy in results being obtained and also consistency in visual clarity.

3 Results and Findings

The visualizations plotted using Python provided clear, data-driven evidence to support the findings of this project. All the analyses performed below can be accessed through this Google Colab [link](#). Below are key takeaways from each of the above listed analyses performed on the dataset:

3.1 Daily Revenue Trend

- The line graph visualizes the daily total revenue (in ₹) generated by the Sudha Milk Parlor over the defined three-month period. It provides a comprehensive view of revenue fluctuations, helping identify performance peaks, dips and recognizable patterns over time.
- The graph displays a strong start in December and early January, with the revenue being relatively high and consistent. Several days during this phase exceeded both the mean and median revenue levels observed within this period, indicating a period of stable demand from the customers. This trend can likely be attributed to the holiday season (with Christmas and New Year) and increased consumer spending. Thus, this period can be capitalized on in the future by stocking the best-selling products in higher quantities.
- The graph also shows a fluctuation in the mid-January period as there is a noticeable increase in volatility, with alternate spikes and dips in the shop's revenue. The exact cause of this cannot be accurately predicted from the dataset alone, since the potential reasons could include stockouts, operational inconsistencies, or variations in the consumer behaviour.
- A steep decline is also evident during the period of mid-to-late February, with the last few weeks of the dataset showing a downward trend, culminating in the lowest revenue day as well. This period requires attention, as it may highlight issues such as declining customer footfall or other external reasons.
- Apart from the sheer analyses of the sales trends, certain key statistical measures can also be calculated from the plot such as the mean revenue and median revenue generated in this period.

- The mean revenue generated by the shop is approximately ₹1824 (marked with a green dashed line in the plot), representing the average earning per day across the 3-month period of the data collection. This serves as a key performance benchmark as days above this line (or value) can be considered as strong performers, while those below suggest scope for improvement.
- Similarly, the median revenue generated by the shop is around ₹1878 (marked with a red dashed line in the plot). The fact that the median is slightly higher than the mean suggests that the presence of several lower outliers (such as the dip in revenue during February) pulling mean revenue down, despite a general trend of moderate-to-good sales on most days in this period.

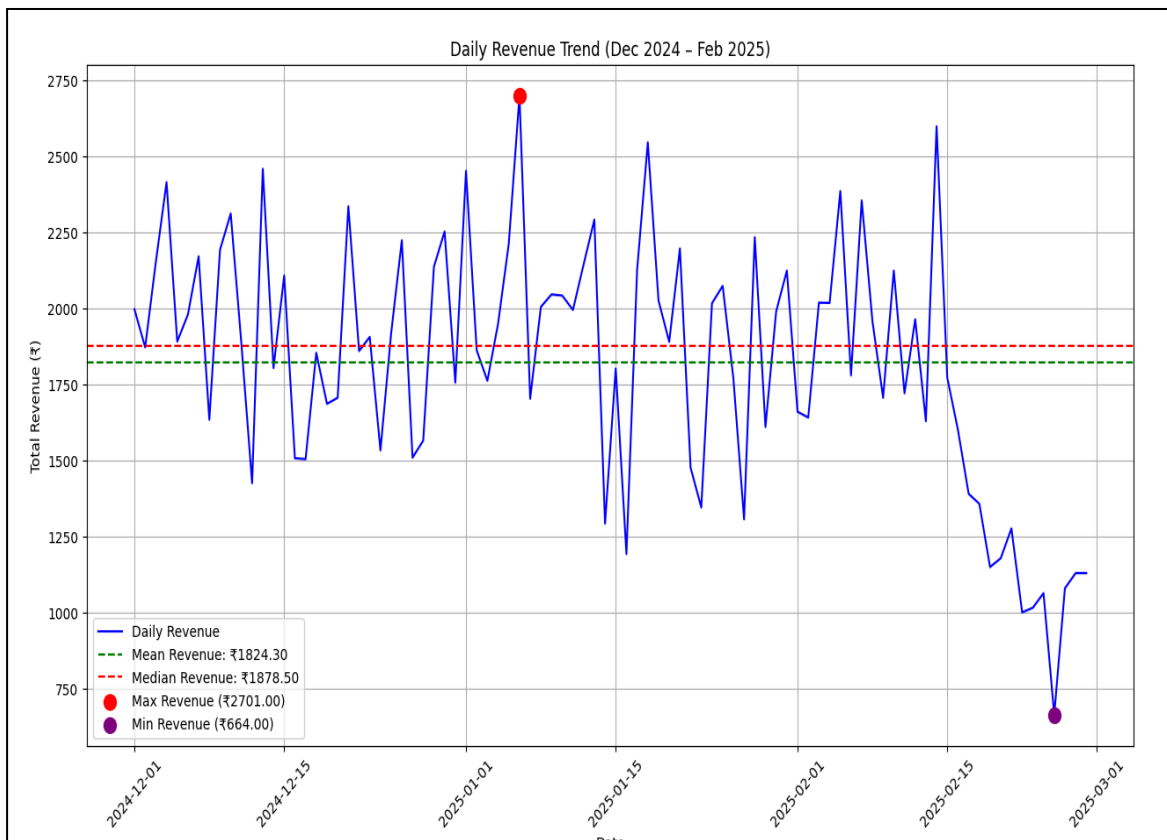


Figure 1: Daily Revenue Trend

3.2 Product-Wise Revenue Contribution

- From the pie chart, it is evident that the 1L variant of Cow Milk was the top contributor to the shop's overall revenue, as it accounted for approximately 34.5% of the total earnings of the shop.
- The 500mL variant of the Sudha Gold Milk followed closely, which contributed around 26.3% of the total revenue, while the 1L variant of the Shakti Milk contributed around 19.4%.

- Collectively, these three products can be considered as the best-selling products as they were responsible for 80.2% of the total revenue of the shop, thus making them the shop's primary revenue sources.
- On the other hand, the 500mL variant of Cow Milk, the 500mL variant of Shakti Milk and the 1L variant of the Sudha Gold Milk were the least-performing products in terms of sales, contributing only about 19.8% of the overall revenue.

This analysis highlights that approximately 80% of the shop's total revenue is generated by just 50% of the products. This provides a clear indication of which products should be prioritized while purchasing, and which ones should be purchased in lesser quantities to optimize inventory space and profitability.

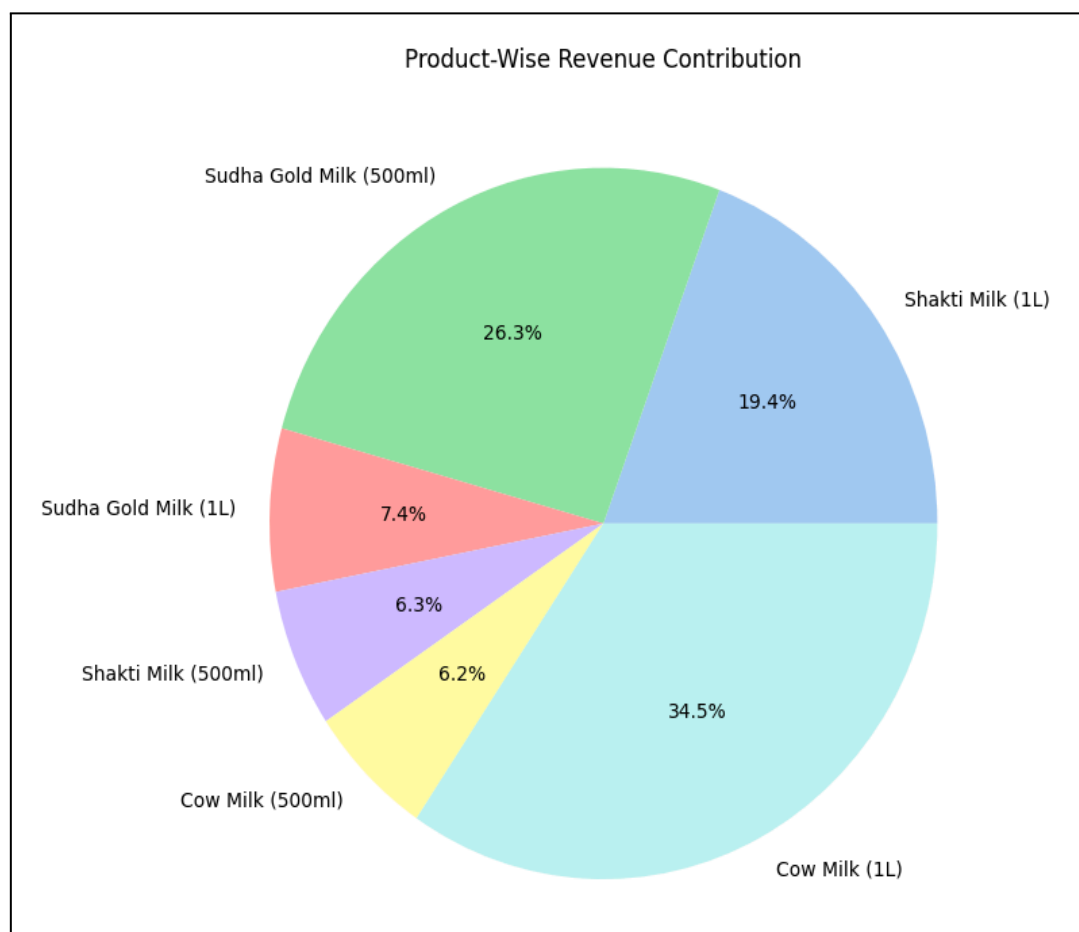


Figure 2: Revenue contribution of individual products

- It is also notable that two out of the three best-selling products are from the 1L variant of the products, indicating their dominance in overall revenue contribution. Interestingly, even the best performer among the slow-moving products, the 1L variant of the Sudha Gold Milk, also belongs to the 1L quantity category.

- Conversely, two out of the three least-selling products are from the 500mL variant of the products, further reinforcing the pattern. This trend may be attributed to the general consumer tendency for purchasing larger quantities, which reduces the frequency of restocking and may also offer better value for money.

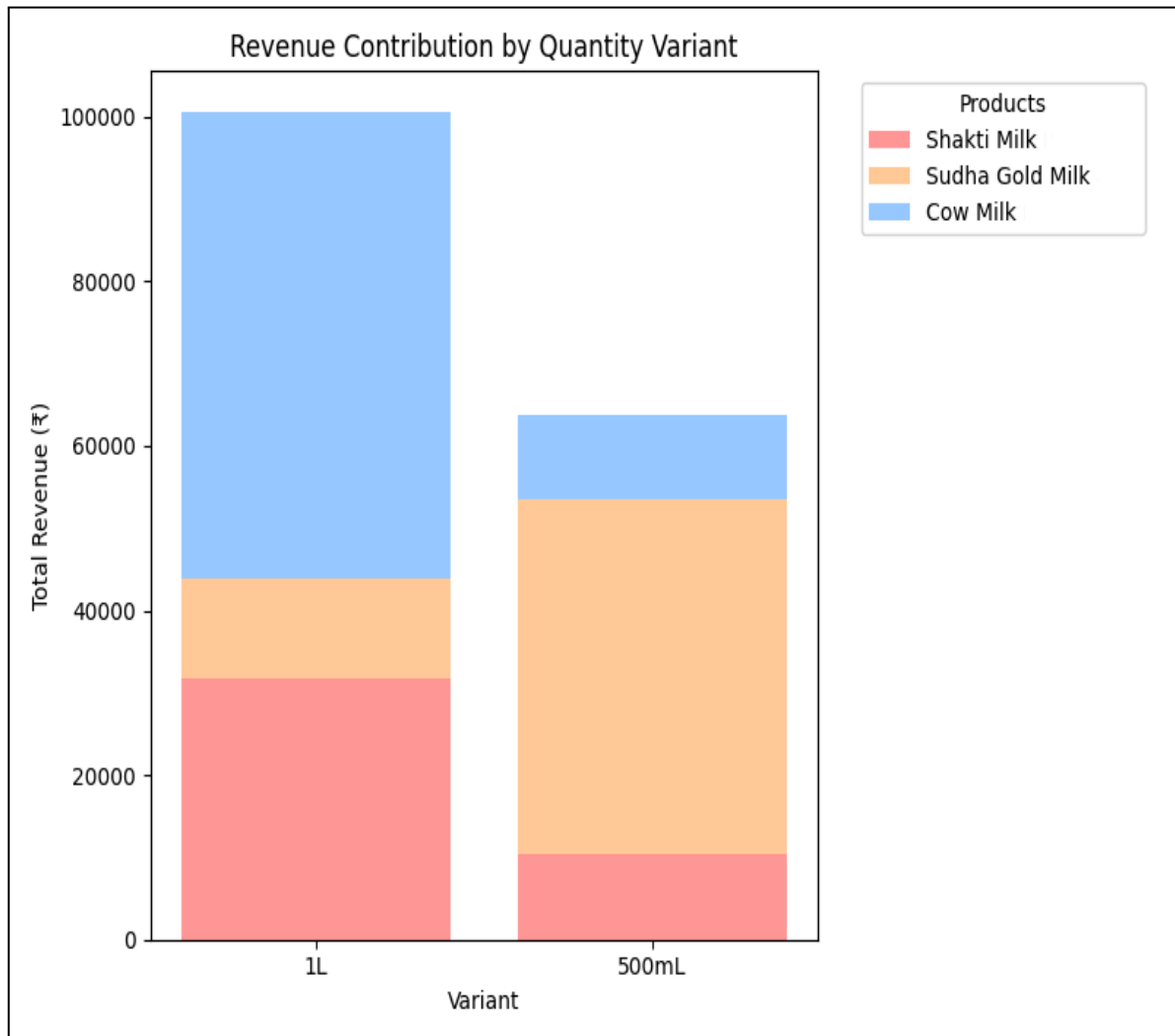


Figure 3: Comparison of Revenue Contribution among Quantity Variants

3.3 Sales Conversions Rates

- The 1L variant of Shakti Milk, the 500mL variant of Sudha Gold Milk, and the 1L variant of Cow Milk recorded sales conversion rates exceeding 99.8%, indicating that nearly all the purchased quantities were sold, with only around 5 units remaining unsold over a period of three months cumulatively. These products can therefore be classified as fast-moving items, highlighting high customer demand and efficient inventory turnover.

- In contrast, the 1L variant of Sudha Gold Milk, the 500mL variant of Shakti Milk, and the 500mL variant of Cow Milk were identified as slow-moving products, with a total of 41 units left unsold during the same period cumulatively. These products not only contribute less to overall profitability but also occupy limited inventory space, leading to financial losses for the shop.

This analysis demonstrates the impact of product movement speed on inventory efficiency and profitability. By identifying and prioritizing fast-moving products, the shop can ensure quicker turnover and reduce the risk of product wastage. On the other hand, minimizing the procurement of slow-moving items can help free up valuable inventory space and improve the shop's overall financial performance.

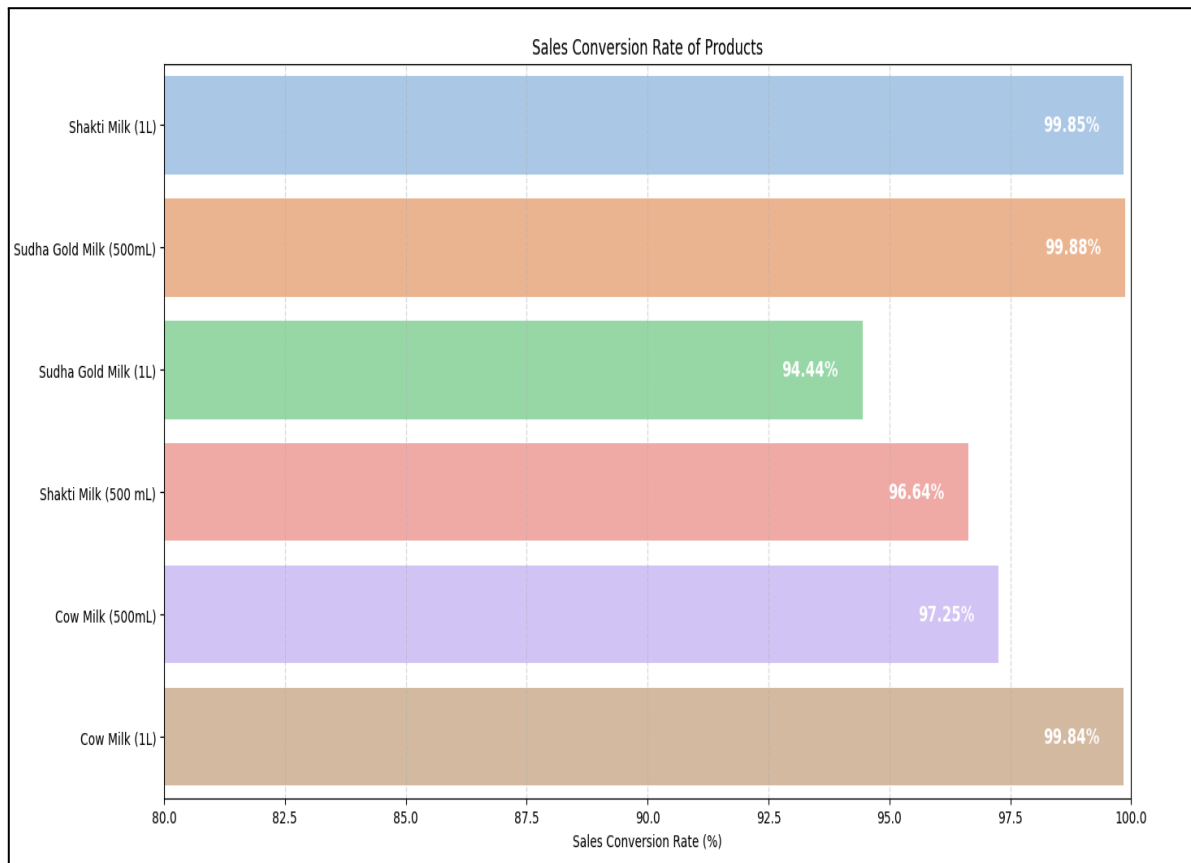


Figure 4: Sales Conversion Rates of individual products

3.4 Profit Margins of Individual Products

- The 500mL variant of the Sudha Gold Milk was the most profitable product for the shop, followed closely by the 1L variant of the Cow Milk.
- On the other hand, the 1L variant of the Sudha Gold Milk caused a net loss to the shop's revenue, while the 500mL variant of the Shakti Milk and the 500mL variant of

the Cow Milk contributed the lowest profit margins, making them the least profitable products that are sold by the shop.

This profitability analysis emphasises the need to focus on products that produce high profit margins while reducing the purchase of products with significantly less or negative profit margins. This analysis is in line with the “Product-Wise Revenue Contribution Analysis” presented in Section 3.1, indicating which products should be purchased more in quantity and which ones should be purchased in lesser quantities.

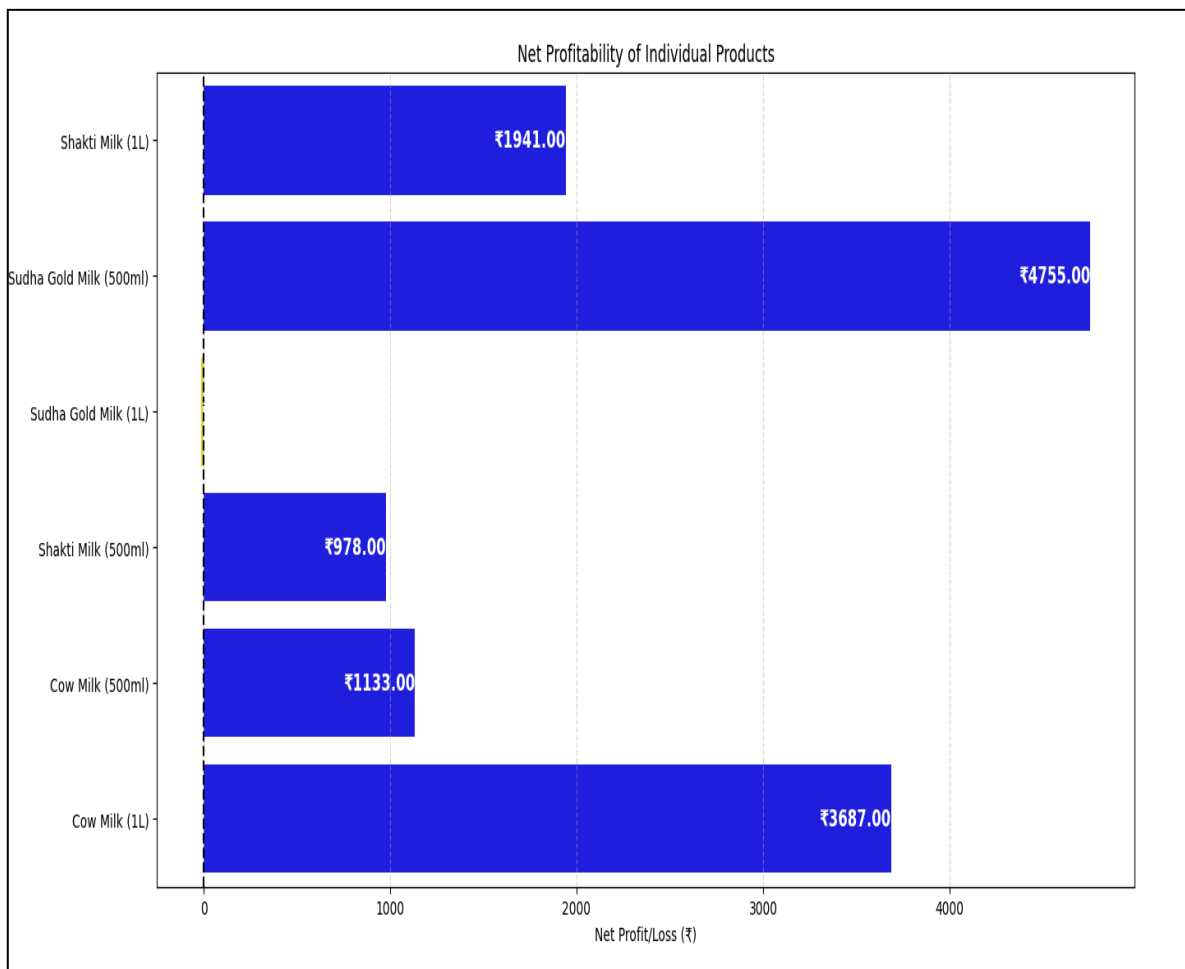


Figure 5: Sales Conversion Rates of individual products

3.5 Optimal Purchase Quantity of Individual Products

- From the bar chart, it is observed that the 500mL variant of the Sudha Gold Milk, the 1L variant of the Shakti Milk and the 1L variant of the Cow Milk were purchased in lesser quantities than their corresponding optimal quantities. Thus, these products

should be purchased in slightly higher quantities than the current quantities to maximize potential profits.

- Conversely, it is observed that the 1L variant of the Sudha Gold Milk, the 500mL variant of the Shakti Milk and the 500mL variant of the Cow Milk were purchased in greater quantities than their corresponding optimal quantities. Thus, reducing the purchase of these products to match their optimal quantities can help in minimizing overstocking and reducing unnecessary inventory costs.

The comparative analysis between actual and optimal purchase quantities highlights that incorporating data-driven purchase decisions, which take into account demand variability, restocking intervals, and conversion rates, can significantly improve the procurement strategy. Following such a strategy will not only reduce instances of understocking and overstocking but also enhance customer satisfaction, improve inventory efficiency, and ultimately increase the profit margins of the shop.

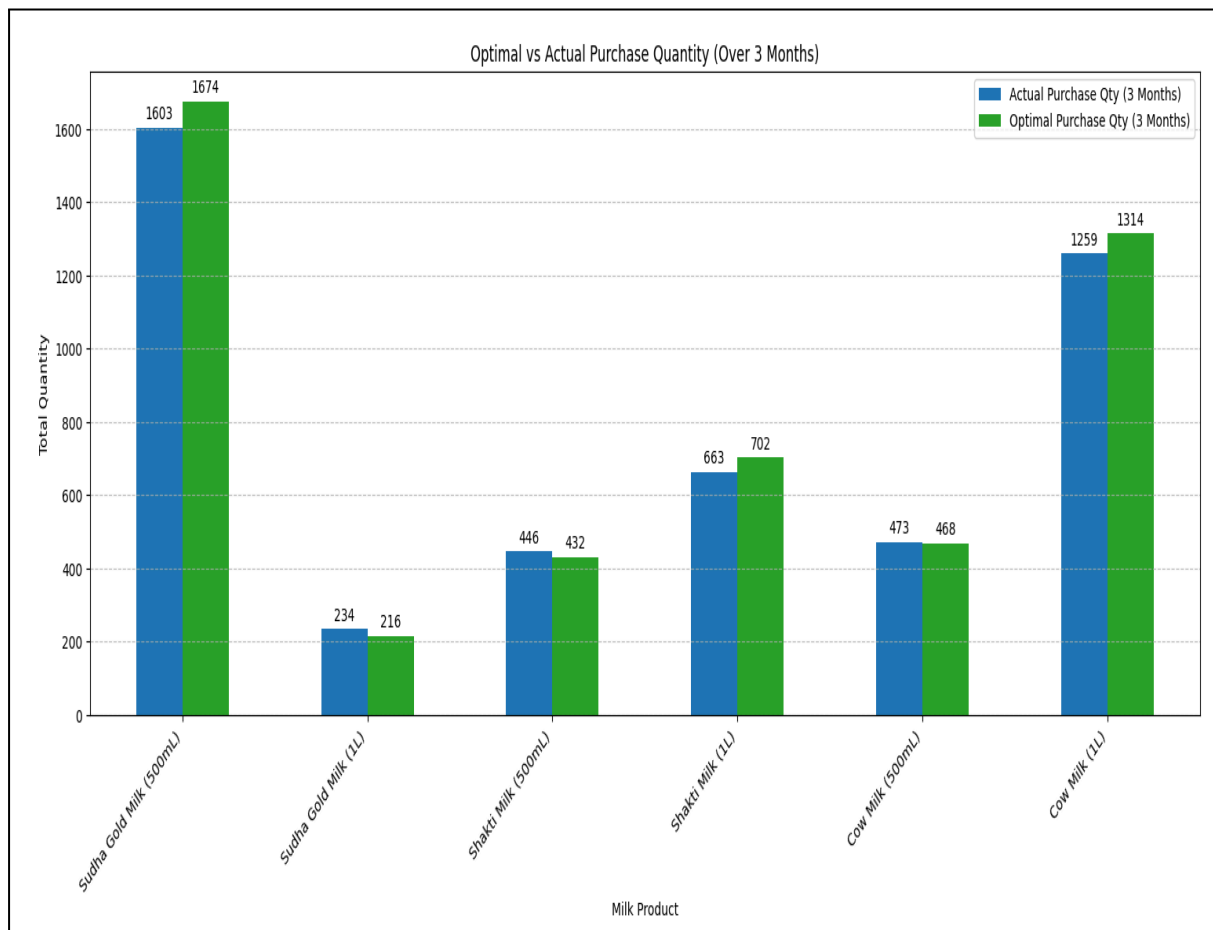


Figure 6: Optimal Purchase Quantities of Individual Products

4 Interpretation of Results and Recommendations

This section contains the detailed interpretation of insights that were obtained from various analyses performed on the dataset and the possible recommendations which can potentially boost the shop's overall performance.

4.1 Interpretation of Results

- The analyses performed on the dataset, considering the product-wise revenue generation, sales conversion rates of individual products and profit margins of each product, revealed notable inefficiencies in inventory management.
- The high-demand products such as the 500mL variant of the Sudha Gold Milk, the 1L variant of the Shakti Milk, and the 1L variant of the Cow Milk were under-purchased compared to their optimal quantities. This indicated potential revenue loss due to understocking.
- Conversely, products such as the 1L variant of the Sudha Gold Milk, the 500mL variant of the Shakti Milk and the 500mL variant of the Cow Milk were overstocked, leading to excess inventory space utilization and possible wastage of products.
- The insights obtained in all the key analyses yielded consistent results and reinforced each other. In conclusion, the 500mL variant of the Sudha Gold Milk, the 1L variant of the Shakti Milk and the 1L variant of the Cow Milk emerged as the best-selling products.
- Conversely, the 1L variant of the Sudha Gold Milk, the 500mL variant of the Shakti Milk and the 500mL variant of the Cow Milk were identified as slow-moving products, highlighting a mismatch between the purchasing decisions and sales performance.
- The sales conversion analysis further revealed varying performance across products, with some products performing significantly below expectations.

These findings collectively emphasize the need for data-driven products procurement strategies and products pricing strategies to enhance inventory efficiency and the shop's financial performance.

4.2 Recommendations

Based on the results obtained from analyses performed on the dataset, a few recommendations are inferred, which when implemented, can potentially improve the revenue generated by the shop and accelerate the shop's growth. The following are the recommendations inferred from the results:

4.2.1 Optimal Purchase Quantity Strategy

- The analysis of sales and purchase data over the three-month period revealed clear mismatches between product demand and procurement quantities. To address this inefficiency, an optimal purchase quantity strategy was developed using average daily sales, sales conversion rates, a 5-day restocking cycle (which is the default restocking frequency used by the shop), and a 5% safety margin.
- This strategy allows the shop to dynamically adjust purchase quantities based on actual sales behavior. Products like the 1L variant of the Shakti Milk, the 500mL variant of the Sudha Gold Milk and the 1L variant of the Cow Milk were under-purchased compared to their optimal restocking requirements, indicating missed potential profitability. Increasing procurement for such high-performing products will help avoid stockouts and ensure consistent availability of products to the customers.
- Conversely, products such as the 1L variant of the Sudha Gold Milk, the 500mL variant of the Shakti Milk and the 500mL variant of the Cow Milk were over-purchased, leading to excess inventory and higher risk of product expiration before sales. Reducing procurement of these items to optimal levels will help minimize product wastage and improve inventory turnover.
- Implementing this optimized procurement strategy would not only improve profitability, but also align product purchasing patterns with the actual customer demand.

4.2.2 Dynamic Pricing Strategy

- One of the major challenges identified in the project was the wastage of unsold products nearing their expiration dates. To tackle this challenge, a dynamic pricing approach is recommended where product prices are adjusted based on their shelf-life status and historical sales performance (sales conversion rates).
- Products with high conversion rates can maintain their existing selling prices or be marginally increased to maximise returns. Meanwhile, slow-moving or expiring products can be offered at discounted prices to stimulate sales and reduce waste.
- Products with a history of low conversion rates or nearing expiration (within 2-3 days) can be discounted in a tiered approach. For example,
 - If a certain product is about to expire in 2 days, then it can be sold at a 25% discounted price
 - If a certain product is about to expire in 1 day, then it can be sold at a 50% discounted price.
 - If a certain product is about to expire on the same day, then it can be sold at a 75% discounted price.
- Fast-moving products retain their original pricing scheme unless nearing expiration (which is quite rare based on the analysis performed in Section 3.3)

- Such a pricing strategy has multiple benefits such as:
 - Helps in balancing inventory levels.
 - Increases the likelihood of selling slow-moving or expiring products.
 - Reduces product being expired before selling.
 - Reduces potential financial losses in the shop.
 - Improves customer perception by offering occasional discounts.

4.2.3 Prioritizing 1L Variants of Milk Products

- The sales and revenue analyses conducted in the project highlighted the better performance of 1L milk variants over their 500mL counterparts.
- Based on this insight, the shop is encouraged to
 - Prioritize procurement of 1L variants for existing and future product lines.
 - Test new products starting with the 1L variant before introducing the smaller variants (say, 500mL)
 - Allocate more shelf space to the 1L variants.

Thus, this strategic focus on the 1L variants can lead to better stock turnover, higher profitability and improved alignment with customer purchasing behaviour.

5 Conclusion

The project demonstrates how systematic analysis of business data can uncover critical inefficiencies in the business and offer actionable insights. It emphasizes the importance of adopting a data-dependent product procurement and inventory management strategy for Sudha Milk Parlor, which can balance product availability and profit optimization.

In the future, further improvement in performance could be achieved by using time-series forecasting for demand prediction, seasonality trend analysis, and potentially applying predictive analytics for more accurate procurement planning strategies. In addition, integrating customer feedback and real-time sales tracking could also help fine-tune product offerings and enhance the overall customer satisfaction.

This project affirms that effective business data management can serve as a powerful tool for small-scale businesses like Sudha Milk Parlor, enabling them to make strategic decisions that can improve operational efficiency and business growth.